MSc Big Data & Finance

Master Thesis — Personal Methodology Notes

A companion to the April 30 intermediate report: what each hypothesis is, what method I used, why I picked it, and what the models actually do

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For personal reference. Not part of the supervisor submission.

1. Why this document exists

The intermediate report I sent to Bobo Zhang and Jian Wu on April 30 is tight, formal, and skips a lot of the reasoning behind why I chose specific methods. That is on purpose: supervisors want to see results and a clean argument, not a tutorial. But three months from now, when I sit down to extend this work for the September submission, I will have forgotten half of why I did things one way and not another. This document is the long version I am writing for myself.

For each hypothesis I cover four things: (1) what the hypothesis is testing in plain English, (2) what method I used, (3) why I used that method specifically and not the obvious alternatives, and (4) what the model is actually doing under the hood.

> **How to use this doc**
> Read this first if you have forgotten what the project is about. Read it alongside intermediate_analysis.py if you want to see exactly which lines of code correspond to which step. Read the intermediate report if you want the polished argument and the numbers.

2. The big picture

The whole thesis sits on one observation from Berg, Kölbel, and Rigobon (2022): ESG ratings from different providers (MSCI, Sustainalytics, Refinitiv, S&P Global) only correlate at around 0.54 with each other. Compare that to credit ratings from Moody's and S&P, which correlate at almost 1.0. So when two professional rating agencies look at the same firm and one calls it green and the other calls it dirty, something is going wrong.

There are two stories about why this happens. The benign story is that sustainability is genuinely multi-dimensional: agencies emphasize different things in good faith. The cynical story is that ESG ratings reward firms for how loudly they talk about sustainability, not for what they actually do. Christensen, Hail, and Leuz (2021) and Drempetic, Klein, and Zwergel (2020) both find evidence consistent with the cynical story: bigger firms with bigger sustainability departments and bigger reports get better ratings, even after controlling for actual environmental and social performance.

My thesis sits on the cynical side and tries to do something about it. The idea: if ESG ratings are partly noise driven by disclosure quality, then I can use machine learning to predict what an ESG rating "should" be from a firm's observable fundamentals (carbon intensity, board structure, controversy history, financials). The gap between the actual rating and the predicted rating is what I call the inconsistency score. If a firm is rated much higher than its fundamentals justify, I treat that as a red flag for greenwashing. If a firm is rated lower than its fundamentals justify, it is potentially underrated. Either way, the corrected score should be a cleaner sustainability signal than the raw rating.

Inconsistency_i = ESG_observed_i − ESG_predicted_i (from ML)

Then for the portfolio side, I tilt a portfolio using the corrected score instead of the raw score and check whether the corrected version produces a better risk-adjusted return. That is the whole project in one paragraph.

3. The four hypotheses, restated with intuition

H1: ML predictive accuracy

The hypothesis: "A machine learning model can predict observed ESG ratings from firm fundamentals with statistically significant out-of-sample accuracy."

Why it matters: if the answer is no, the entire thesis falls apart. If fundamentals do not predict ESG ratings at all, then either the rating agencies are reading something I cannot observe (which means my "fundamentals-based" prediction is meaningless), or ratings are pure noise. Either case kills the inconsistency-score idea before it starts. So H1 is a sanity check on the entire premise. I expect to reject $H1_0$ comfortably; if I do not, I have a problem.

H2: drivers of the inconsistency score

The hypothesis: "Firms with high positive inconsistency scores have higher disclosure intensity, larger size, and cluster systematically by sector."

Why it matters: H1 tells me there IS a gap between observed ESG and what fundamentals would predict. H2 asks whether that gap looks like greenwashing or like noise. If the gap is concentrated in big firms with fancy sustainability reports, it is consistent with the disclosure-driven ratings story. If the gap is random, the inconsistency score might just be model error, not a meaningful signal.

H3: portfolio performance

The hypothesis: "ESG-tilted portfolios using inconsistency-adjusted scores reach higher out-of-sample Sharpe ratios than portfolios using raw ESG scores."

Why it matters: H1 and H2 are about understanding the data; H3 is about whether the understanding is worth anything in practice. Anyone managing money under an ESG mandate is forced to use ratings that are at least partly noise. If my correction produces a better signal, that has direct economic value. If it does not, the thesis is intellectually interesting but practically irrelevant.

H4: signal-purity robustness (auxiliary)

The hypothesis: "The inconsistency-adjusted ESG score has lower cross-provider rating disagreement than the raw ESG score."

Why it matters: this is an indirect validation. If my correction works, the corrected scores from multiple providers should agree more with each other than the raw scores do, because the correction strips out the provider-specific noise. It is auxiliary because I can only run it properly with two or more provider feeds, which I do not have yet. For the pilot I substitute a within-method version (correlation across the three ML model specs).

4. The data pipeline

Everything flows through a single Python script (intermediate_analysis.py) that runs end-to-end in roughly two minutes. The pipeline is deterministic: np.random.seed(42) is set at the top, so re-running gives identical numbers. The flow:

- Load master_dataset.csv (100 firms × 25 columns) from eda_outputs/. Drop the two 100%-missing columns. Median-impute small holes in derived ratios.
- Build a 25-feature design matrix: 15 numerical features + 10 sector dummies (11 sectors, drop one to avoid collinearity).
- H1: train Ridge / Random Forest / XGBoost with 5-fold CV. Compute out-of-sample R^2, RMSE, MAE. Run a 200-replicate permutation test on the best model.
- H1 SHAP: refit the best model on the full sample and compute SHAP values for feature attribution.
- H2: compute residuals from the best model = inconsistency score. Standardize. Run an OLS regression of standardized inconsistency on candidate drivers + sector dummies, with HC1-robust standard errors. Compute quintile means.
- H3: simulate 36 months of monthly returns using each firm's observed beta + a market factor + idiosyncratic noise + an ESG-pricing kernel. Build three z-score-tilted portfolios. Compute Sharpe / vol / drawdown / Calmar. Run Jobson-Korkie test on Sharpe difference.
- H4: compute Pearson and Spearman correlations of residuals across the three ML specifications.
- Save everything: 9 CSV files, 5 PNG figures, 1 JSON summary. The Word doc reads from those files, so the report and the pipeline cannot drift.

5. H1 deep dive: ML predictive accuracy

5.1 What I am actually predicting

The target variable is the Sustainalytics ESG total score, a continuous number between roughly 5 and 60 where lower is better (low risk = good). The predictors are 25 features:

- Size and scale: log market cap, revenue, employees, total debt, EBITDA.
- Profitability: net income, ROE, ROA, profit margin.
- Leverage and liquidity: debt-to-equity, current ratio.
- Risk and valuation: beta, P/E ratio, P/B ratio.
- ESG-relevant: controversy indicator (0–5 ordinal scale).
- Sector: 10 one-hot dummies (Communication Services is the dropped reference category).

Notice that I am explicitly NOT including the E, S, or G sub-scores as features. That would be circular: the sub-scores partly determine the total ESG score by construction. The whole point is to predict the rating from objective firm characteristics, not from other ratings.

5.2 Three models, three philosophies

Ridge regression (the linear baseline)

Ridge is just OLS regression with a penalty on coefficient size. The objective function is:

$$\min_\beta \; \Sigma (y_i - x_i'\beta)^2 + \lambda \Sigma \beta_j^2$$

The $\lambda \Sigma \beta_j^2$ term shrinks coefficients towards zero. Why bother? Two reasons. First, with 25 features and 100 firms, plain OLS would overfit; Ridge regularizes that away. Second, when features are correlated (and many of mine are: revenue and market cap, ROE and ROA), OLS produces unstable coefficients that swing wildly between samples; Ridge stabilizes them. The cost: coefficients are biased towards zero, so you cannot read them as causal.

I choose λ via nested cross-validation: an outer 5-fold CV evaluates performance, and an inner CV inside each training fold picks the best λ from 25 candidates between 0.001 and 1000. This avoids data leakage; the model never sees the test fold during hyperparameter selection.

Ridge is fast, interpretable, and a strong baseline. It is also what wins on this pilot, which surprised me; see Section 5.3.

Random Forest (the bagging ensemble)

A random forest is 500 decision trees, each trained on a bootstrap sample of the data, each split using only a random subset of features. Predictions are averaged across all 500 trees.

The intuition: a single decision tree is high-variance (small data changes produce very different trees). Averaging many decorrelated trees cancels out the variance without much bias cost. Bagging + random feature selection is what makes the trees decorrelated.

Random forests are good at non-linearities and interactions, robust to outliers, and handle mixed feature types without scaling. The cost: they need enough data per tree to learn meaningful structure. With 100 firms and 5-fold CV, each fold trains on 80 observations. That is borderline thin for trees with depth 8.

My hyperparameters: 500 trees, max depth 8, min samples per leaf 3. I did NOT tune these via grid search for the pilot — I picked sensible defaults from the literature. This is something I want to revisit when the panel is bigger.

XGBoost (the boosting ensemble)

Gradient boosting is fundamentally different from bagging. Instead of training many trees in parallel and averaging, you train trees sequentially, where each new tree fits the residuals (errors) from the previous trees. After T rounds, the prediction is the sum of all T tree contributions.

$$F_t(x) = F_{t-1}(x) + \eta \cdot h_t(x), \quad \text{where } h_t(x) \approx -\partial L/\partial F_{t-1}(x)$$

XGBoost adds two key engineering tricks on top of plain gradient boosting: (1) it uses a second-order Taylor expansion of the loss function for faster convergence, and (2) it includes built-in L1 and L2 regularization on the tree weights to control overfitting. Chen and Guestrin (2016) made it the de facto standard for tabular ML.

My hyperparameters: 400 boosting rounds, learning rate 0.05 (slow learner = better generalization), max depth 4 (shallow trees = less overfit), subsample 0.8 (each tree sees 80% of rows = stochastic gradient boost), colsample 0.8 (each tree sees 80% of features), reg_alpha 0.1, reg_lambda 1.0. These are reasonable defaults. Bayesian optimization is on the to-do list for the panel extension.

5.3 Why Ridge wins (and why that is not a problem)

Out-of-sample R^2: Ridge 0.493, XGBoost 0.378, Random Forest 0.310. The linear baseline beats both non-linear models. Initially I thought I had a bug, but the result is real and explainable.

It comes down to the bias-variance tradeoff. Tree ensembles have low bias and high variance: they can capture complex patterns, but they need a lot of data to do so reliably. Linear models have higher bias and lower variance: they cannot capture non-linearities, but they are stable. With only 100 firms, the variance term dominates. Ridge's simplicity becomes an asset, not a limitation.

Gu, Kelly, and Xiu (2020) found XGBoost dominant on a panel of ~25,000 firm-months. I have 100 firm-years. Of course the comparison flips. The fix is more data, not more model. When the panel reaches ~1,800 firm-year observations, I expect XGBoost to take the lead, because the variance issue shrinks as N grows.

5.4 Why 5-fold CV (and not the time-series CV from the methodology)

The methodology section of the preliminary work specifies time-series CV with an expanding window: train on 2015 only, test on 2016; train on 2015-2016, test on 2017; and so on. This prevents look-ahead bias because the model never sees future data during training. It is the right approach for a panel.

But my pilot dataset is cross-sectional (one snapshot in time, not a panel), so time-series CV does not apply. I use plain 5-fold CV with shuffling and random_state=42 instead. When the panel arrives, I will switch back to time-series CV. The intermediate report flags this as a methodology refinement.

5.5 The permutation test (and why I run one)

A high R^2 could be a real signal or a fluke from random chance, especially with 25 features and 100 observations. I want a clean significance test on whether $R^2 > 0$ is real.

The permutation test: I randomly shuffle the y vector (so the relationship between fundamentals and ESG is broken), refit the same model with 5-fold CV, and record the resulting R^2. Repeat 200 times. The result is an empirical null distribution of R^2 under "no relationship". The p-value is the fraction of the null distribution that exceeds my actual R^2. For Ridge, none of the 200 permuted runs got close to $R^2 = 0.493$, so $p < 0.005$. The signal is real.

> **Why permutation and not an F-test**
>
> I could compute a regression F-test on the Ridge fit, but the F-test assumes i.i.d. Gaussian errors, which my residuals do not satisfy (they are heteroskedastic by sector). Permutation tests are non-parametric and assume only that observations are exchangeable under the null. They are slower (200 refits ≈ 30 seconds) but more honest in finite-sample non-Gaussian settings.

5.6 SHAP values (and why I trust them)

A model can predict ESG well without me understanding which features it is using. SHAP (SHapley Additive exPlanations) solves this. For each prediction and each feature, SHAP attributes a numerical contribution: how much did this feature push the prediction up or down relative to the baseline (mean prediction)?

The trick: SHAP values come from cooperative game theory. They are the unique attribution that satisfies four desirable properties (efficiency, symmetry, dummy, additivity). Lundberg and Lee (2017) showed how to compute them efficiently for tree models in particular.

For my Ridge model, SHAP values reduce to scaled coefficients (because Ridge is linear), but I still use the SHAP framework for consistency: when I switch to XGBoost on the panel, the same code will produce comparable feature attributions. On the current pilot, SHAP confirms what I expect: sector membership and operating profitability (ROA) are the dominant predictors of ESG ratings.

6. H2 deep dive: drivers of the inconsistency score

6.1 Why OLS for this step (not another ML model)

For H1 I want maximum predictive accuracy, so I throw the whole ML toolkit at it. For H2, I want interpretable coefficients with confidence intervals — I want to know, for each candidate driver, whether the relationship with inconsistency is statistically significant and what its sign and magnitude are. ML models are bad at this; OLS is purpose-built for it.

So the workflow is two-stage: ML for prediction (H1), OLS for inference (H2). This is the standard pattern in applied econometric ML papers. Mullainathan and Spiess (2017) call this "prediction policy problems" and explain why the two stages serve different goals and use different tools.

6.2 What HC1-robust standard errors are and why I use them

In OLS, the standard error of each coefficient depends on the assumption that the residuals are homoskedastic (constant variance). With financial data, they almost never are: bigger firms have bigger residuals, and certain sectors have wider error variance than others. If I ignore this and use standard OLS errors, my p-values will be wrong (usually too small, leading to overconfident significance claims).

Heteroskedasticity-Consistent (HC) standard errors fix this. White (1980) introduced HC0; Davidson and MacKinnon (1993) introduced refinements HC1 through HC4. HC1 is the most common choice for financial cross-sections — it includes a small-sample correction factor n/(n−k) that prevents the errors from being too small in finite samples.

$$\mathrm{Var\_HC1}(\beta) = (n / (n-k)) \cdot (X'X)^{-1} X' \Omega X (X'X)^{-1}$$

where Ω is the diagonal matrix of squared OLS residuals. Reading: replace the homoskedastic $\sigma^2 I$ with the empirical residual variance, scaled for degrees of freedom. The point estimates do not change; only the standard errors do.

6.3 Choice of candidate drivers

The candidate drivers come from H2 itself: disclosure intensity, size, sector, and a few financial controls. My specific choices:

- log_market_cap as the size proxy. Drempetic, Klein, and Zwergel (2020) show larger firms get systematically higher ESG ratings; if size positively predicts inconsistency, that supports the disclosure-driven ratings story.
- controversy as a direct ESG-related driver. Firms with higher controversy scores should logically have lower ESG ratings; if they instead have positive inconsistency (rated higher than fundamentals predict), that suggests disclosure or PR is offsetting controversies in the rating.

- debt_equity, roa, profit_margin, pe_ratio as financial controls. These are standard panel controls in asset pricing; they account for cross-sectional variation that is not really "ESG" but might be correlated with it.
- 10 sector dummies. The H2 sub-question is whether inconsistency clusters by sector; this directly tests it.

Notably absent: a direct disclosure-intensity proxy (number of pages in the sustainability report, number of GRI indicators reported, etc.). On the pilot I did not have time to scrape this. For the panel extension I plan to use the Bloomberg ESG Disclosure Score as a proxy.

6.4 What the regression actually shows

Sector dummies dominate. Five out of ten are individually significant: Utilities ($p = 0.001$), Industrials ($p = 0.002$), Health Care ($p = 0.003$), Financials ($p = 0.024$), Information Technology ($p = 0.017$). All are negatively signed relative to Communication Services (the reference). Reading: firms in these sectors have systematically lower inconsistency than Communication Services firms, holding fundamentals constant. So Communication Services firms appear to be over-rated by Sustainalytics on this sample, or other sectors are under-rated.

The size and controversy effects are directionally consistent with H2 but not statistically significant in this sample. This is a power issue, not a rejection: with 100 firms and 16 right-hand-side variables, the standard errors on individual coefficients are wide.

The full regression has F-statistic $p < 0.001$, so jointly the model has significant explanatory power. $R^2 = 0.184$, adjusted $R^2 = 0.027$. About 18% of inconsistency variation is explained by these candidate drivers; the rest is idiosyncratic noise + measurement error + omitted drivers (e.g. disclosure intensity, which I do not have on the pilot).

7. H3 deep dive: portfolio backtest

7.1 The z-score tilt method

I want to compare three portfolios that are identical except for the ESG signal they use. The standard approach in quant ESG papers is the z-score tilt around an equal-weight or cap-weight benchmark.

$$w_i = w_i^{benchmark} + \gamma \cdot z(ESG_i) / N$$

Where w_i^benchmark is the benchmark weight (1/N for equal-weight), z(ESG_i) is the cross-sectionally standardized ESG score for firm i, γ controls the tilt intensity (I use $\gamma = 0.5$), and N is the number of firms. Then I clip negative weights to zero and renormalize so the weights sum to 1.

The tilt magnitude is small ($\gamma = 0.5$, divided by $N = 100$, so each firm's weight moves by at most a few tenths of a percent for typical z-scores). This keeps the portfolios diversified and ensures performance differences come from the ESG signal, not from concentration changes.

7.2 The three portfolios

- Benchmark. Equal-weighted, no ESG tilt. This is the do-nothing baseline.
- Raw-ESG. z-score tilt on the raw Sustainalytics ESG total score. This is what a standard quant ESG manager would do today.
- Adjusted-ESG. z-score tilt on the inconsistency-corrected score: ESG_corrected_i = ESG_observed_i – α × Inconsistency_i, with $\alpha = 1$. This is my proposed signal.

The α parameter controls how much I penalize the inconsistency. $\alpha = 1$ fully subtracts the inconsistency, yielding the pure ML-prediction. $\alpha = 0.5$ keeps half the original signal. $\alpha = 0$ reduces the Adjusted portfolio to the Raw portfolio. For the pilot I use $\alpha = 1$ as the baseline and run robustness checks across {0.5, 1.0, 1.5, 2.0} for the panel extension.

7.3 Why a synthetic backtest (and what makes it valid)

My pilot dataset is cross-sectional. I have one ESG snapshot per firm. To compute portfolio Sharpe ratios I need a return time-series, which I do not have on the pilot. Two options: skip the H3 test entirely, or simulate returns calibrated to known empirical relationships and treat the result as a methodology validation rather than an economic test.

I went with simulation. The motivation is that I want to confirm the entire pipeline works (z-score tilt → portfolio weights → return series → performance metrics → Jobson-Korkie test) before plugging in real data. Bugs in the methodology are easier to catch in a controlled simulation than in a real backtest.

7.4 The return-generating process

Each firm's monthly excess return is generated as:

$$r_{\{i,t\}} = rf/12 + \beta\_i \cdot (r\_m,t - rf)/12 + esg\_kernel\_i / 12 + \varepsilon\_\{i,t\}$$

where rf = 4% (risk-free rate, annualized), r_m,t is a market factor drawn monthly from N(8%/12, (16%/√12)²), β_i is the firm's observed beta from the data, esg_kernel_i = −1.5% × inconsistency_z (i.e. firms with high positive inconsistency lose 150 bps per year of expected excess return), and ε_{i,t} is i.i.d. idiosyncratic noise N(0, (18%/√12)²).

The 1.5% calibration of the ESG kernel comes from Pástor, Stambaugh, and Taylor (2021), who model in equilibrium how an ESG quality premium depends on investor preferences and information asymmetry, and who calibrate the premium to roughly 1-2% per year for typical preferences. I use the conservative end of that range.

Critically, the ESG kernel is built into the return-generating process before the portfolio-construction step. The Adjusted-ESG portfolio sees a clean signal (it tilts on inconsistency-corrected scores, which match the kernel by construction). The Raw-ESG portfolio sees a noisier signal (raw ESG = fundamentals + inconsistency, but the kernel only prices inconsistency). So in expectation, the Adjusted portfolio should outperform.

7.5 The performance metrics, and what each one captures

- Annualized return. Just (1 + monthly mean)^12 – 1. Tells you if you made money on average. Says nothing about risk.
- Annualized volatility. Monthly std × √12. Tells you how rough the ride was. Says nothing about asymmetry (drawdowns can be much worse than vol suggests).
- Sharpe ratio. (Annualized return – risk-free rate) / annualized volatility. The single most-used risk-adjusted return measure. Higher is better. Sharpe = 1.0 is good, 2.0 is great, 3.0 is suspicious.
- Maximum drawdown. The worst peak-to-trough loss observed. Captures left-tail risk that vol misses. A portfolio with low vol can still have a big drawdown if the bad months cluster.
- Calmar ratio. Annualized return / absolute max drawdown. A "Sharpe ratio for tail risk". Higher = more return per unit of worst-case pain.

7.6 The Jobson-Korkie test

Two portfolios both have Sharpe = 2.15. Is the difference of 0.003 statistically significant or just noise? The Jobson-Korkie (1981) test answers this directly. It computes the asymptotic standard error of the Sharpe difference, accounting for the correlation between the two return series, and gives you a z-statistic.

$$z = (\hat{S}_1 - \hat{S}_2) / \sqrt{[\ (1/n) \cdot (2 - 2\rho + \tfrac{1}{2}(\hat{S}_1^2 + \hat{S}_2^2 - 2\hat{S}_1\hat{S}_2\rho^2))\ ]}$$

For my Adjusted vs Raw comparison: $z = 0.367$, $p = 0.71$. The two Sharpes are statistically indistinguishable. Reading: with $N = 100$ firms and 36 months, I do not have enough data to detect a Sharpe difference of 0.003 even if it exists. I expect the panel extension with real returns to give a much cleaner verdict, in either direction.

Why I am not panicking about p = 0.71

A non-rejection is not the same as a confirmed null. The pilot was always a methodology validation, not a definitive economic test. The framework runs, the test statistic is well-defined, the numbers are reproducible. The actual economic question waits on real return data, which is the priority for the final thesis.

8. H4 deep dive: signal-purity robustness

8.1 What H4 was supposed to test (and why I cannot test it directly)

The original H4 says: if my correction works, the inconsistency-corrected ESG scores from MSCI, Sustainalytics, Refinitiv, and S&P Global should agree more with each other than the raw scores do. Operationally: compute the standard deviation of corrected scores across providers for each firm, and compare it to the standard deviation of raw scores across providers. If the correction is removing provider-specific noise, the corrected std should be smaller.

The blocker: Yahoo Finance only exposes Sustainalytics. To run the real H4 test I need at least one additional provider. The two viable options for academic access are MSCI ESG Research and Refinitiv Eikon, both via WRDS (the Wharton Research Data Services subscription that the school holds). I have flagged this as the top data-access priority. Until then, H4 has to be tested indirectly.

8.2 The within-method substitute

If the inconsistency score reflects a real economic phenomenon — not just noise in one specific estimator — then the residuals from Ridge, Random Forest, and XGBoost should agree on which firms look over-rated and which look under-rated. This is a weaker version of the cross-provider test but shares the spirit: ask whether the signal is robust to specification choices.

I report two correlations:

- Pearson correlation of residual values. Captures how closely the magnitudes of inconsistency agree between models. Sensitive to the shape of the joint distribution.
- Spearman rank correlation. Captures how closely the rankings agree. Robust to monotonic transformations and outliers. If the question is "does Ridge and XGBoost agree on which firms are most over-rated?" Spearman is the right measure.

On my pilot, Pearson correlations are 0.84 to 0.95, Spearman 0.85 to 0.95, with the two non-linear models in particularly tight agreement. The signal is robust to the choice of estimator. That is consistent with H4 in spirit, though it is not the cross-provider test that H4 originally specified.

9. Honest limitations summary

Things I should remember when reading my own results:

- N = 100 is small. Most results are directionally meaningful but underpowered for small effect sizes. The panel extension is a serious priority.
- Cross-sectional, not panel. I cannot use time-series CV, cannot do real backtests, cannot study how ratings drift over time. The synthetic backtest compensates partially for H3 but it is not a substitute for real data.

- Single provider. Yahoo Finance / Sustainalytics is the only ESG source. H4 is fundamentally about cross-provider disagreement; without a second source, the headline H4 test is impossible.
- Disclosure intensity is missing. The H2 prediction depends on disclosure-driven ratings, but I do not have a direct disclosure proxy on the pilot. Bloomberg's ESG Disclosure Score is the planned addition.
- Synthetic backtest is calibrated. The 150 bps ESG kernel I bake into the return-generating process is reasonable but not data-driven. A real backtest could give very different results.
- No regime analysis. ESG performance varies a lot across periods (Friede et al. 2015, Dimson et al. 2020). My single-period cross-section cannot say anything about regime dependence.
- Median imputation is crude. For the pilot I used per-variable median imputation, which ignores sector and time-series structure. Sector-by-year imputation is the planned upgrade.

10. Quick glossary

- R^2 (coefficient of determination): proportion of target variance explained by the model. 1.0 = perfect, 0.0 = no better than predicting the mean.
- RMSE (root mean squared error): typical prediction error in target units. ESG units in my case. Penalizes big errors more than small ones.
- MAE (mean absolute error): average absolute prediction error. Same units as target. Less sensitive to outliers than RMSE.
- Out-of-sample / OOS: evaluated on data the model did not see during training. This is what generalizes; in-sample fit is misleading.
- Ridge regression: OLS with an L2 penalty. Shrinks coefficients to control overfitting.
- Random Forest: ensemble of decision trees trained on bootstrap samples, predictions averaged. Bagging reduces variance.
- XGBoost: gradient boosting library. Trees trained sequentially on residuals. Industry-standard for tabular ML.
- SHAP (SHapley Additive exPlanations): per-prediction feature attribution based on cooperative game theory. Gives you "how much did each feature contribute to this specific prediction".
- OLS (ordinary least squares): standard linear regression. Minimizes squared residuals.
- HC1: heteroskedasticity-consistent standard errors with a small-sample correction. Use when residual variance is non-constant.
- Sharpe ratio: (excess return) / volatility. Risk-adjusted return measure.
- Calmar ratio: annualized return / max drawdown. Tail-risk-adjusted return measure.
- Jobson-Korkie test: significance test for the difference between two Sharpe ratios.
- Permutation test: non-parametric significance test that builds the null distribution by shuffling the target variable many times.
- z-score / standardization: subtract mean, divide by std. Puts a variable on a unit-variance, zero-mean scale.
- Inconsistency score: ESG_observed minus ESG_predicted (from ML). Positive = over-rated relative to fundamentals; negative = under-rated.